

Deliverable 3.2

Description of the System Architecture of the platform, including the composing, the intelligent and the blockchain components

Project Name: **Digitalising European Uncontested Claims Enforcement**

Grant Agreement No.: **101138437**

Call: **JUST-2023-JCOO**

Start date of the project: **1 April 2024**

Duration of the project: **24 months**



DEUCE Consortium Disclaimer

This document contains material, which is the copyright of certain DEUCE contractors, and may not be reproduced or copied without permission. All DEUCE consortium partners have agreed to the full publication of this document. The commercial use of any information contained in this document may require a license from the proprietor of that information. The reproduction of this document or of parts of it requires an agreement with the proprietor of that information. The document must be referenced if used in a publication. This report has been produced with the financial support of the JUST Programme of the European Union. The contents of this report are the sole responsibility of the authors and can in no way be taken to reflect the views of the European Commission.

The DEUCE consortium consists of the following partners:

No.	Role	Short Name	Legal Name	Country
1	COO	VUB	VRIJE UNIVERSITEIT BRUSSEL	BE
2	BEN	UL	UNIVERZA V LJUBLJANI	SI
3	BEN	VU	VILNIAUS UNIVERSITETAS	LT
4	BEN	UniZg	SVEUCILISTE U ZAGREBU – PRAVNI FAKULTET	HR
5	BEN	UNINA	UNIVERSITA DEGLI STUDI DI NAPOLI FEDERICO II	IT
6	BEN	UB	UNIVERSITAT DE BARCELONA	ES
7	BEN	TalTech	TALLINNA TEHNIKAÜLIKOO	EE
8	BEN	Adiconsum	ASSOCIAZIONE ITALIANA DIFESA CONSUMATORI ED AMBIENTE	IT
9	BEN	EUBF	FONDATION DE LA CHAMBRE EUROPEENNE DES HUISSIERS DE JUSTICE	BE
10	BEN	FBE	FEDERATION DES BARREAUX D'EUROPE	FR
11	BEN	ZEV	ZENTRUM FÜR EUROPAISCHEN VERBRAUCHERSCHUTZ EV	DE

Document Information

Project short name and number	DEUCE (101138437)
Work package	3
Number	D3.2
Title	System Architecture of the platform for monetary claim procedure
Type ¹	R
Dissemination level ²	SEN
Contractual date of delivery	31.03.2025

Document History

Version	Date	Status	Authors, Reviewers	Description
v 0.01	25.03.2024	Template	A: Gioia Arnone [VUB]	Project deliverable template
v 0.02	15.11.2024	First Draft	A: Flora Amato, Alberto Moccardi, Rajib Chandra Ghosh [UNINA] R: Marco Giacalone [VUB]	First draft. Document structure and content.
v 0.03	17.12.2024	Draft	A: Flora Amato, Alberto Moccardi, Francesco Moscato, [UNINA]	Revision of the work.
v. 0.04	11.01.2025	Draft	A: A: Flora Amato, Alberto Moccardi, Rajib Chandra Ghosh [UNINA] R: Marco Giacalone,	Revision of the work.

¹**Types.R:** Document, report (excluding the periodic and final reports); **DEM:** Demonstrator, pilot, prototype, plan designs; **DEC:** Websites, patents filing, press & media actions, videos, etc.; **OTHER:** Software, technical diagram, etc.

²**Dissemination levels:** Public — fully open (e.g. web); Sensitive — limited under the conditions of the Grant Agreement; EU classified — RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision [2015/444](#).



Co-funded by
the European Union

'Funded by the European Union. Views and opinions expressed are however those of the author (s) only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the granting authority can be held responsible for them.'

			Susanna Picariello [VUB]	
v 0.05	20.02.2025	Draft	A: A: Flora Amato, Francesco Moscato, Rajib Chandra Ghosh [UNINA]	Revision of the work.
v 0.06	14.03.2025	Draft	A: Alberto Moccardi, Mattia Fonisto, Rajib Chandra Ghosh [UNINA] R: Marco Giacalone, Susanna Picariello [VUB]	Revision of the work.
v 1.0	16.03.2025	Final Draft	A: Flora Amato [UNINA] R: Marco Giacalone [VUB]	Final Revision
	26.03.2025	Final Draft	A: Flora Amato [UNINA] R: Artem Bojarchuk, Oleksii Khramov, Thomas Hoffmann [TalTech] Marco Giacalone, Susanna Picariello [VUB]	Harmonizing formatting issues, adding blockchain contribution

Table of Content

I. Background to the DEUCE project.....	7
II. Scope of the Document.....	7
III. Relationship to Other Project Deliverables	8
IV. Structure of the Document.....	8
V. Modal verbs (RFC 2119).....	9
1. Introduction to Platform Design.....	10
1.1 Overall Platform Requirements	10
1.2 Software Architecture (AI).....	11
2. Core Logical Schema	16
2.1 Compose Component.....	17
2.2 Intelligent Component.....	17
3. Integrated Component Interactions.....	19
3.1 Workflow Orchestration and Intelligent Processing	19
3.2 Document Management and Verification Workflow	20
3.3 Interim conclusion	20
4. Immutable Audit Trail System Using Permissioned Blockchain.....	21
4.1 Introduction	21
4.2 Architectural Integration	21
4.3 Technical Design of the Permissioned Blockchain.....	22
4.4 Integration Points with Existing System Components.....	24
4.5 GDPR Compliance and Data Protection.....	25
4.6 Compliance and Regulatory Benefits	27
4.7 Implementation Approach.....	27
4.8 Query and Reporting Capabilities	28
4.9 Future Directions	28
4.10 Summary	28

List of figures

Fig. 1: Overview of the Software Architecture main blocks (SA)	12
Fig. 2: Schematic detailed representation of the Software Architecture (SA)	15
Fig. 3: Internal workflow of the two logical components	16
Fig. 4: Architecture of the Intelligent Interface	18
Fig. 4.1: Blockchain System Architecture showing integration layers with existing components	21
Fig. 4.2: Byzantine Fault Tolerant consensus flow showing the pre-prepare, prepare, and commit phases	22
Fig. 4.3: Detailed view of block structure and transaction composition	23
Fig. 4.4: Workflow showing how case lifecycle events are recorded to the blockchain	24
Fig. 4.5: Hybrid storage model showing on-chain vs. off-chain data storage with pseudonymization	25

Definitions, acronyms and abbreviations

- DEUCE - Digitising European Uncontested Claims Enforcement;
- EEO - European Enforcement Order;
- EOP - European Order for Payment;
- UML – Unified Modelling Language;
- API – Application Program Interface;
- AI - Artificial Intelligence
- ML - Machine Learning
- BPM – Business Process Management
- DMS – Document Management System
- DSS - Decision Support System
- SA – Software Architecture

This Project has received funding from the European Commission JUST 2027 Programme under grant agreement no. 101046587. This document has been prepared for the European Commission, however it reflects the views only of the authors, and the Commission cannot be held responsible for any use which may be made of the information contained therein.

I. Background to the DEUCE project

DEUCE will rely on the successful results obtained within the previous three projects managed by the Consortium: SCAN (2018-2020, JUST-JCOO-AG-2017), EFFORTS (2020-2022, JUST-JCOO-AG-2019) and SCAN II (2018-2021, JUST-JCOO-AG-2017). In its activities, DEUCE will significantly benefit from SCAN II and EFFORTS databases (the former for ESCP Regulation and the latter for EEO and EOP guidelines), from SCAN experience for what concerns surveys and interviews for the creation of the comprehensive Roadmap as well as from the strong and supportive network of various stakeholders (i.e., ECC-Nets, legal practitioners, academics). From a methodological perspective, the synergic collaboration among the legal unit partners in conducting a plethora of research activities and hands-on tasks is demonstrated by the successful outputs achieved in the above-mentioned projects. Furthermore, DEUCE's objectives complement and support the EU-LISA widespread efforts to digitalize EU Justice. Currently the front-runner Agency in managing large-scale IT systems, EU-LISA's core activity consists of interoperability, namely the efficient communication and cooperation of IT systems to provide EU law enforcement officials with quicker access to comprehensive information and to improve cross border judicial cooperation. Hence, by promoting the use of EEO and EOP procedures, which offer simplified instruments for the recovery of monetary claims, DEUCE will enable the development of a favorable environment for the large-scale adoption of the e-CODEX Project, now being incorporated into the EU-LISA framework. The envisaged result thus consists of the implementation of an interoperable, secure and decentralised communication network between national IT systems in cross-border civil and criminal proceedings.

II. Scope of the Document

The scope of this document is to provide a comprehensive description of the system architecture for the platform dedicated to European monetary claim procedures. This document particularly outlines the key architectural components and their interdependencies, focusing on the integration of data processing and blockchain.

In detail, the document covers the following aspects:

- **Platform Components Overview:**
A high-level description of each primary module, including the Intelligent Processing Module, Blockchain Store, Document Management System, Client Application, and External Integration Interfaces.
- **Detailed Architecture and Software Infrastructure:**
An in-depth explanation of the internal structure and functionality of the processing components data analytics, decision support, and risk assessment including how these components work together to process claim data, validate transactions, and maintain an audit trail.
- **Interfaces and Communication Protocols:**
A detailed description of the communication mechanisms and APIs that interconnect the different system modules and data flows between the fundamental block of the architecture.
- **Intelligent Data Processing:**
An overview of the methodology deployed in the intelligent interface as an instrument for legal and users to gain useful information about claim and regulations.

It is important to note that this document does not mention the complete implementation details of individual components. Instead, it focuses on the architectural integration, identifying the

components and outlining how each contribute to the overall functionality ensuring functional and non-functional requirements.

III. Relationship to Other Project Deliverables

This document builds on and complements previous deliverables by providing an architectural perspective that integrates the technical software design and the business process management (BPM) approach for monetary claim procedures under a requirement setting. The current deliverable extends and aligns the software components with the workflow processes, ensuring that every technical element is designed to support and integrate seamlessly with the end-to-end claim resolution process.

Deliverable 3.1 focused on establishing a standardized workflow using BPM techniques (e.g., BPMN) and automating the procedural steps in handling monetary claims. The proposed document references and incorporates the BPM flow insights by mapping to the corresponding technical components. This ensures that the intelligent processing modules and blockchain infrastructure directly support and enhance the automated workflow defined in the BPM deliverable but do not present any technical details and technological stack.

IV. Structure of the Document

The document is organized in a logical progression that begins with establishing the contextual foundation by introducing the background of the project and providing an overview of the objectives and the relevance of previous work. After laying out the background, the document defines its scope, clarifying the boundaries of the architectural discussion and specifying what aspects of the system architecture are covered. It then examines how the current document builds on earlier software and BPM-related outputs to create a cohesive view of the platform. Following these initial sections, the document depicts a high-level architectural overview describing the fundamental design principles that underpin the system's macrostructure and detailing how various components interact at a broad level. The subsequent chapters focus on the platform's core subsystems dedicated to the composing component, explaining its role in orchestrating data flows, managing concurrency, and integrating external microservices or specialised backends and the intelligent component, illustrating how advanced algorithms, heuristic methods, and machine learning models NLP can be integrated to ensure robust performance under varying workloads. The document then examines how these subsystems collectively operate, detailing the integration mechanisms that allow external entities to interact with the system while maintaining strict security and performance standards. Reaching the end, a discussion of security, privacy, and regulatory compliance, emphasizing how the architectural elements are aligned with legal and operational guidelines, is proposed, also highlighting possible deployment strategies and operational concerns, including environment configuration and runtime monitoring, concluding with an analysis of potential maintenance challenges, future enhancements, and areas for refinement.

- ❖ **Context and Background:** Introduces the project, its objectives, and prior work that underpins this deliverable.
- ❖ **Scope and Relationship:** Defines the boundaries of the document and explains how it integrates with previous deliverables.

- ❖ **High-Level Architectural Overview:** Outlines the fundamental technologies and design principles that shape the system's overall structure.
- ❖ **Subsystem Analysis:** Examines the core components, including the composing component for data orchestration and the intelligent component for advanced analytics.
- ❖ **Integration:** Describes how external interfaces, security measures, and regulatory guidelines are incorporated.

V. Modal verbs (RFC 2119)

1. **MUST**—This word, or the terms “REQUIRED” or “SHALL”, means that the definition is an absolute requirement of the specification.
2. **MUST NOT**—This phrase, or the phrase “SHALL NOT”, means that the definition is an absolute prohibition of the specification.
3. **SHOULD**—This word, or the adjective “RECOMMENDED”, means that there may exist valid reasons in particular circumstances to ignore a particular item.
4. **SHOULD NOT**—This phrase, or the phrase “NOT RECOMMENDED” means that there may exist valid reasons in particular circumstances when the behaviour is acceptable or even useful, but the full implications should be understood, and the case carefully weighed before implementing any behaviour described with this label.
5. **MAY**—This word, or the adjective “OPTIONAL”, means that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product, while another vendor may omit the same item.

1. Introduction to Platform Design

The platform for monetary claim procedures is engineered as a modular, secure, and scalable system that seamlessly integrates advanced intelligent processing, possibly integrating a robust blockchain-based infrastructure designed to automate and optimise claim resolution processes while upholding the highest standards of data integrity, transparency, and regulatory compliance. By dividing the system into discrete, interoperable components—such as the intelligent processing module, blockchain ledger, document management system, and client interface—the platform achieves a high degree of flexibility, allowing for the rapid integration of new functionalities and easy maintenance. Nonetheless, Security is a foundational aspect of the design, with the blockchain integration recording all transactions and documents immutably through hashing procedures. Moreover, the system's scalable, cloud-native architecture enables horizontal expansion to accommodate growing transaction volumes and user demand without compromising performance and possibly respecting any requirement set. Advanced intelligent processing, powered by artificial intelligence (AI) and machine learning algorithms (ML), facilitates real-time data analysis and automated decision support, possibly through the integration of a specific chatbot that leverages Natural Language Processing (NLP) [1] and Large Language Models (LLMs) [2 , 3] as its main building blocks, streamlining the evaluation and resolution of claims and potentially optimising overall operational efficiency from user and court perspective. Through this comprehensive approach, the platform meets stringent standards and ensures accountability of the Business Process Management (BPM), paving the way for future trustworthy adaptability in an evolving regulatory environment.

1.1 Overall Platform Requirements

As a user-friendly web application, the consolidated platform streamlines the user experience from initiation of claim submission to final resolution.

At its most basic level, the system should be able to automate routine tasks while facilitating real-time decision support to simplify complicated evaluations of user intentions. The functional requirements include several vital areas that together form the platform's operational integrity and usability, which are essential to the software design as they define the form of the solution.

First, the platform should enable the automation of claims handling, including claim submission and registration, methodical tracking, and processing. It should automatically route claims through pre-defined workflows, verify claim eligibility, and trigger necessary actions, such as notifications or escalations.

Secondly, to ensure data integrity and auditability, the system should implement a blockchain-based system to record every transaction, sign documents, and process actions, providing a safe audit trail, which is critical to maintaining trust and complying with regulatory standards. All interactions in the system must be logged so that the history of every claim is openly available for verification and future auditing purposes.

Another fundamental requirement is the combination of sophisticated, intelligent processing with the platform, which must leverage ML, NLP, and LLMs for sophisticated data analytics to streamline the user experience and flow while facilitating real-time risk analysis and automated decision support [4] to classify claims, identify trends, anomalies, and suggest proper actions to the claimant or more generally to the user. The intelligent processing module enhances the evaluation process while, in parallel, offering insightful information that can be utilised to improve and refine the end-to-end claim management process. Moreover, by integrating a robust document management system with the blockchain ledger [6], the solution must not only enhance the authenticity of the documents but also position itself as a key player in the claims process. In

conclusion, the system intends to provide interoperability by providing a rich set of APIs and integration mechanisms enabling seamless interfaces with external systems, identity-proofing services, and public key infrastructures, making the monolithic application more potentially executable. From a practical standpoint, the platform comes with its integration to ensure that it is connectable with all existing external systems available in the market and becomes affordable in terms of the centralization of maximum data related to the various points. In summary, the platform's functional requirements mainly include automatic claim processing, data integrity and audibility, real-time intelligent processing, secure document management, and data interoperability, collectively creating the basis for an effective, efficient, and transparent system that can deal with the complexities in today's digital era.

1.2 Software Architecture (AI)

The overall platform architecture, presented in Fig. 1, is structured to provide a robust and scalable environment for managing monetary claims. It integrates advanced technical components and a comprehensive Business Process Management (BPM) framework. At its core, the system is designed to automate the end-to-end lifecycle of claim processing while ensuring that every step adheres to technical and regulatory standards.

The platform comprises several key modules, each playing a distinct role in supporting the monetary claim procedure. The intelligent processing module leverages data analytics and ML to evaluate claim submissions in real-time, providing decision support that automates routine assessments and flags potential issues. In parallel, the blockchain-based infrastructure should serve as an immutable ledger, recording all transactions and digital signatures to guarantee data integrity and maintain a transparent audit trail.

A secure Document Management System (DMS) is another critical pillar of the architecture, ensuring all claim-related documentation is stored and verifiable. This component works in tandem with the blockchain, reinforcing the overall trustworthiness of the claims process.

The first building block of the software chain client application serves as the primary interface for various stakeholders, including claimants, financial institutions, and legal professionals. It facilitates the submission, tracking, and management of claims while providing real-time notifications and dashboards that reflect the system's intelligent processing outputs.

Central to this architecture is the BPM process, which orchestrates activities across these modules. The BPM framework standardises and automates the tasks involved in claim processing—from initial submission and validation to final resolution and enforcement. By integrating BPM principles, the platform ensures that each step is executed consistently and in accordance with predefined business rules and regulatory requirements. This not only enhances operational efficiency but also provides transparency and accountability throughout the claim lifecycle.

In summary, the overall architecture seamlessly blends advanced technical functionalities with a structured BPM process, creating a secure, scalable, and adaptable system. It supports automated decision-making, robust data integrity, and efficient workflow management, all of which are essential for the reliable and transparent resolution of monetary claims.

1.2.1 Decision Support System

The Intelligent Processing Module is the analytical heart of the platform, designed to process real-time data and support informed decision-making throughout the claim lifecycle. It continuously collects claim and transaction data from various sources, ensuring that the system has access to

the most current information. This data undergoes an extensive preprocessing phase, cleansed, normalised, and transformed (tokenised for LLMs inference) into actionable insights.

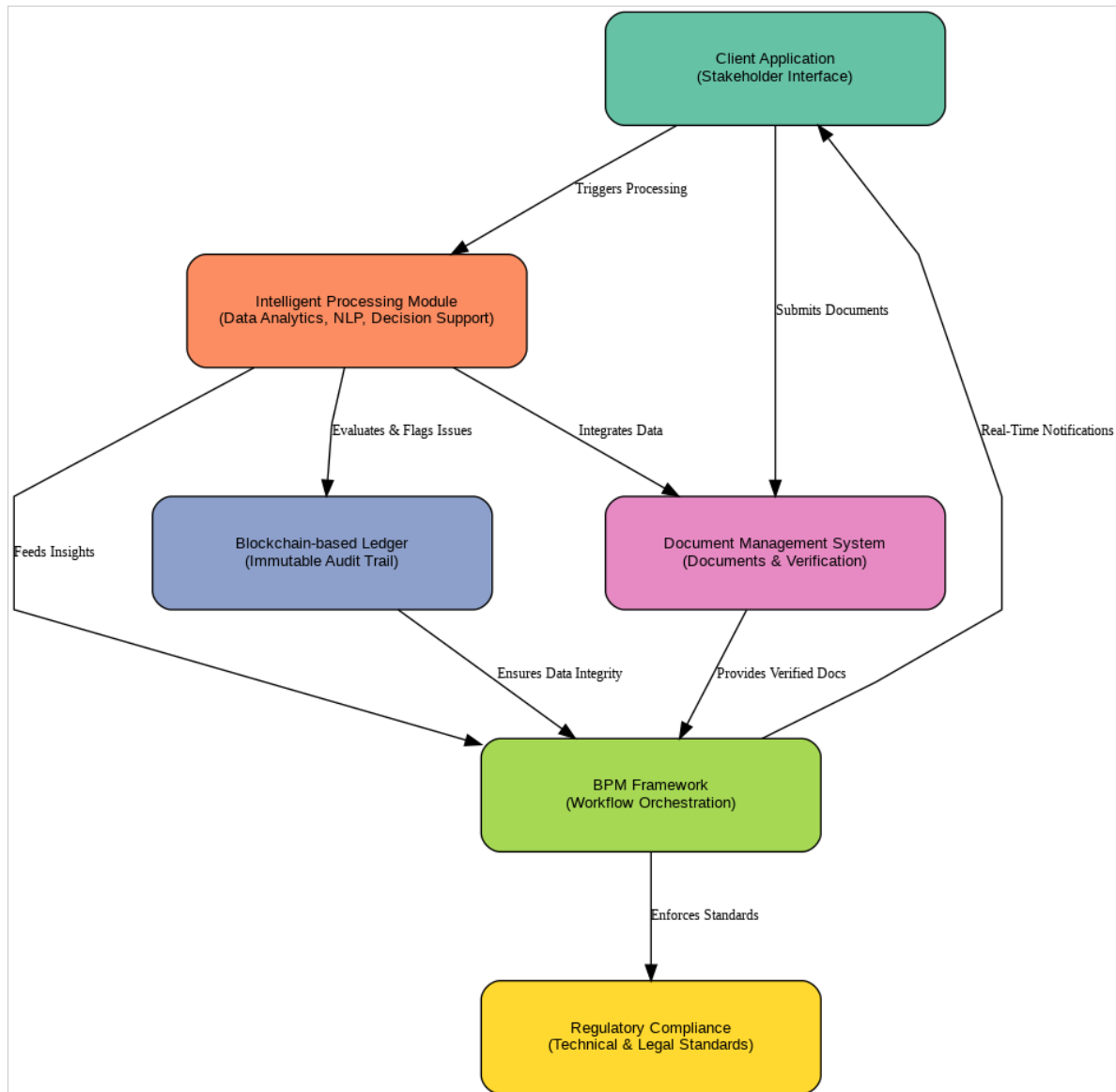


Fig. 1: Overview of the Software Architecture main blocks (SA)

At its ground, this pivotal module applies advanced ML algorithms to analyse the preprocessed data, enabling it to detect patterns, classify different claim types, and predict potential processing times and outcomes based on a set of stored past cases. The core of this module is a robust Decision Support System (DSS) that integrates a business engine that applies predefined legal and procedural rules consistently across all claims, ensuring compliance with regulatory standards [7]. Based on the previous analysis, the recommendation engine automatically suggests the appropriate course of action and potentially recommends the most suitable procedure. To ensure seamless

integration with the broader system, the DSS should be connected to both the blockchain-based ledger and the Document Management System via a suite of APIs, allowing for efficient data exchange and ensuring that all decisions and actions are securely recorded in an immutable ledger notifying alerts of any significant events or breaches [8]. This method clearly facilitates prompt intervention while enhancing operational efficiency and reinforcing the platform's commitment to transparency, security, and regulatory compliance.

1.2.2 Document Management System

The Document Management System (DMS) is engineered as a secure and resilient repository for all claim-related documentation designed to ensure the confidentiality, integrity, and availability of digital documents, possibly through robust encryption techniques and comprehensive access controls. The DMS is equipped with a strong suite of integration APIs to facilitate seamless operations with the rest of the platform and external services. These interfaces allow for secure, standardised, and real-time access to document data, ensuring CRUD modality. From a practical perspective, the DMS supports various data exchange formats and protocols and is equipped with API keys for access control, providing flexibility to integrate with legacy systems, third-party applications, and future technological advancements.

1.2.3 Client Application

The Client Application is designed to serve as a user-friendly interface for various stakeholders, including claimants, financial institutions, legal professionals, and administrative personnel. This module is central to ensuring that all interactions with the platform are intuitive, secure, and efficient through robust user management and authentication role-based access control, ensuring that users only have access to functionalities pertinent to their responsibilities and adopting secure login procedures. In addition, the application facilitates comprehensive claim submission and tracking with the users. It can monitor their progress through a streamlined workflow and possibly inform them of any developments or required actions throughout the claim lifecycle when needed. An interactive dashboard bridging the gap between complex backend processes and user interactions, ensuring that the platform remains accessible, secure, and responsive to the needs of all its users.

At its core, the ClientApp consists of several key components:

- ❖ Core Module: Handles essential functionalities like authentication, HTTP block and route guards.
- ❖ Feature Modules: Implement specific functionalities such as user management, claim submission, document verification, and notification handling.
- ❖ Routing Module: Manages navigation within the application.
- Services: Provide communication with backend APIs and manage application state.

The client module technically interfaces with various backend services through an API Gateway, enabling functionalities like user authentication, document management, blockchain integration for verification, and notifications. It employs responsive design principles to ensure usability across different devices and screen sizes.

1.2.4 External Integration and Communication

The presented platform, designed to operate within a broader ecosystem, may necessitate a seamless interaction with a range of external systems and services made possible via standardised communication channels that ensure efficient data exchange and interoperability.

Additionally, the platform should integrate external Public Key Infrastructure (PKI) services to manage and verify digital signatures. This is essential for maintaining the authenticity and non-repudiation of electronic documents, as it allows for robust verification processes that are consistent with industry standards, enhancing its overall security posture and ensuring that all digital interactions are trustworthy.

Practically, the use of RESTful APIs is central to the platform's external communication strategy. It enables seamless interoperability with third-party applications and systems, allows for standardised data exchange in real-time, simplifies the integration with legacy systems and new technologies, and ensures that the platform remains adaptable to evolving industry standards and emerging requirements.

In conclusion, the platform's external integration and communication framework is built to support secure financial interactions, robust digital signature verification, and flexible, standardised connectivity with third-party systems, thereby ensuring comprehensive interoperability and compliance across the entire ecosystem.

1.2.5 Detailed Architectural Overview

In detail, the entire system is conceived to address the need for a cohesive, end-to-end platform that manages intricate legal, administrative, and intelligent processes while remaining adaptable to future requirements, linking together all their singular component into a monolithic application. To achieve this, one objective of the current solution is the seamless orchestration, whereby critical workflows—such as case initiation, review, and escalation—run smoothly without manual synchronisation overhead while allowing modular extensibility, meaning the platform should allow new features or subsystems to be integrated with minimal rework or disruption to existing components.

This approach is particularly vital in legal and administrative contexts, where regulatory updates or evolving best practices may demand periodic functionality or process logic changes. A related principle is fault-tolerant reliability. Given the significance of compliance with strict legal deadlines and data integrity mandates, the system must remain available and recover gracefully from potential failures or network disruptions. Consequently, high-availability mechanisms—like load balancing, active-active clustering, or automated rollback strategies—should be embedded at the core of the design.

Emphasis should also be placed on the clear separation of concerns among the platform's major subsystems and their codebases. The composing component aims to orchestrate data flows and service interactions, while the intelligent component applies advanced computation techniques to improve decision-making.

This decoupled model simplifies maintenance and testing and allows for independent scaling, ensuring that spikes in resource-intensive data analytics do not degrade overall system responsiveness.

A detailed overview of which extends the Fig.1 also investigating the possible internal flow of each component is depicted in Fig. 2

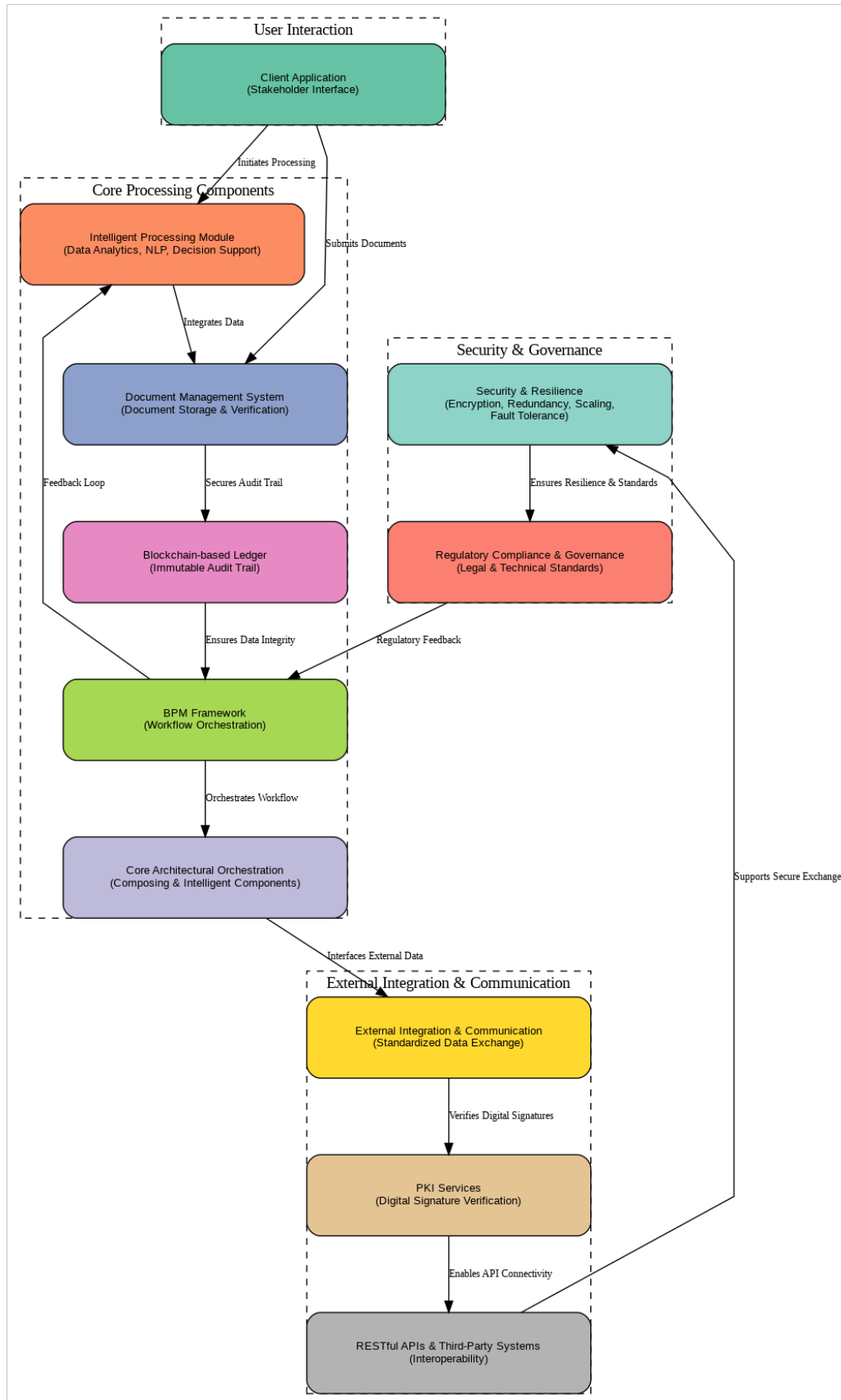


Fig. 2: Schematic detailed representation of the Software Architecture (SA)

2. Core Logical Schema

Internally, the Composing Component is often constructed around a BPM (Business Process Management) engine that interprets process definitions written in standards such as BPMN (Business Process Model and Notation). Each high-level process is represented as a network of tasks, decision points, and parallel executions, with the BPM engine handling runtime execution and transitioning each case through these tasks according to the logic embedded in the process definitions.

Data typically enters the component through secure RESTful API endpoints or event-driven messaging channels, depending on whether the communication is synchronous or asynchronous. Once received, inputs are routed to the relevant workflow instance. Then the workflow engine coordinates any calls to external services or advanced computational modules, tracking both success and failure states. Whenever human interaction is required, the system updates relevant queues and user dashboards, attaching context (case references, task instructions, deadlines) to provide a meaningful work package. Upon task completion, the engine records the result in an audit trail and signals subsequent steps.

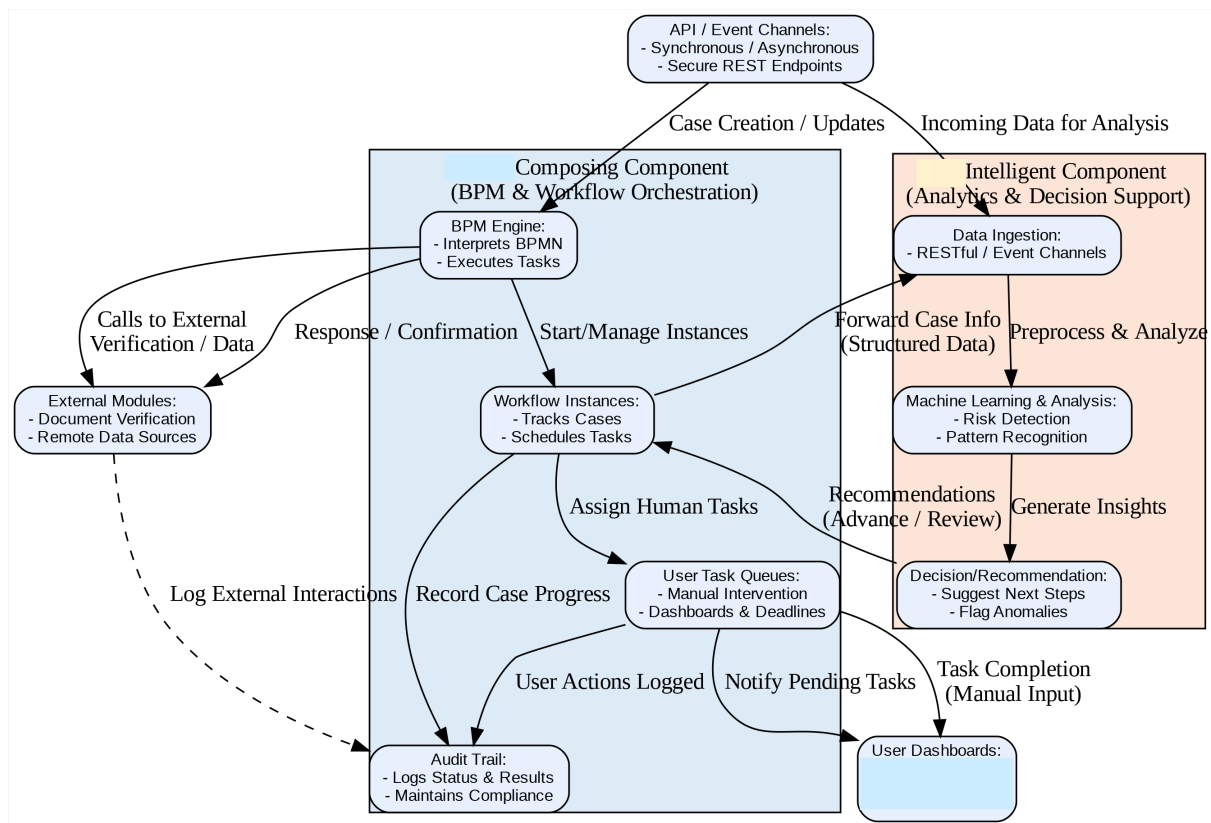


Fig. 3: Internal workflow of the two logical components

This flow, presented in Fig. 3, often follows a request-response pattern with the orchestration layer. Then, the orchestration layer evaluates the current state of the case against predefined

business rules or past cases guided by ML or NLP models, either advancing it automatically to the next step or alerting a user if manual intervention is needed. All these design choices emphasise modularity, enabling the insertion or replacement of sub-flows (e.g., to accommodate revised legal procedures) while retaining the overarching process control logic.

2.1 Compose Component

The Composing Component serves as the platform's backbone to manage procedural workflows by synchronising interactions among various services while ensuring that every submitted request—a claim creation or a document update—flows seamlessly through the appropriate channels. The internal structure encapsulates a triggering logic orchestration following the rules and constraints mandated by legal or administrative procedures, guaranteeing a cohesive end-to-end experience. Practically, this component's importance is heightened when a single user action triggers parallel or sequential tasks involving several subsystems. For example, when a claimant files a case, the Composing Component coordinates between form validation, intelligent processing and advice generation.

2.2 Intelligent Component

The Intelligent Component delivers analytic and decision-support capabilities by processing diverse inputs—from textual case descriptions to structured metadata such as jurisdiction and case type—to produce actionable insights. Building on this foundation, the enhanced intelligent interface resolves economic claims by leveraging state-of-the-art natural language processing and machine learning techniques in a human-in-the-loop framework [9].

At its core, the system employs advanced transformer models that utilise attention mechanisms to interpret user queries accurately. These models are trained on a comprehensive, domain-specific knowledge base comprising legislative texts, case studies, and best practices in dispute resolution. In parallel, an ontology-based model formalises legal expert knowledge using OWL/RDF and may ensure that the system not only comprehends the language and structure of legal documents but also understands the underlying legal relationships and precedents.

The human-in-the-loop integration is a critical feature of the intelligent interface since it incorporates annotation tool technology to extract metadata from legal corpora, supporting continuous improvement and model validation against real case studies leveraging and a curated dataset, ensuring that the models remain both current and relevant. Additionally, an interactive feedback mechanism allows legal professionals to review and adjust the model's performances, triggering a claimed action from the IT team and ensuring that automated decisions and recommendations align with the latest legal standards and practices.

Designed with cross-border cases in mind, the system offers multilingual capabilities by seamlessly linking queries with legal corpora in different languages through the Multilingual NLP model approach, which interprets user queries in one language and offers corresponding legal information and advice in another, aiding in the harmonization of different legal frameworks. This functionality provides tailored responses that account for jurisdiction and legal terminology variations.

The intelligent interface, depicted in Fig. 4, practically continuously analyses incoming data to generate dynamic recommendations, such as suggesting optimal next steps for contested claims based on historical patterns or flagging anomalies in submitted documentation. Its output feeds directly into the workflow engine, ensuring that decision support—whether through automated suggestions for claim progression or alerts for manual intervention—is immediately reflected in the case processing workflow, enhancing overall operational efficiency.

By integrating deep learning models, ontology-based reasoning, and human oversight this advanced, context-aware system streamlines the resolution process for economic claims, adapting

to the complexities of cross-border legal queries while significantly improving the efficiency and reliability of the overall claim resolution workflow.

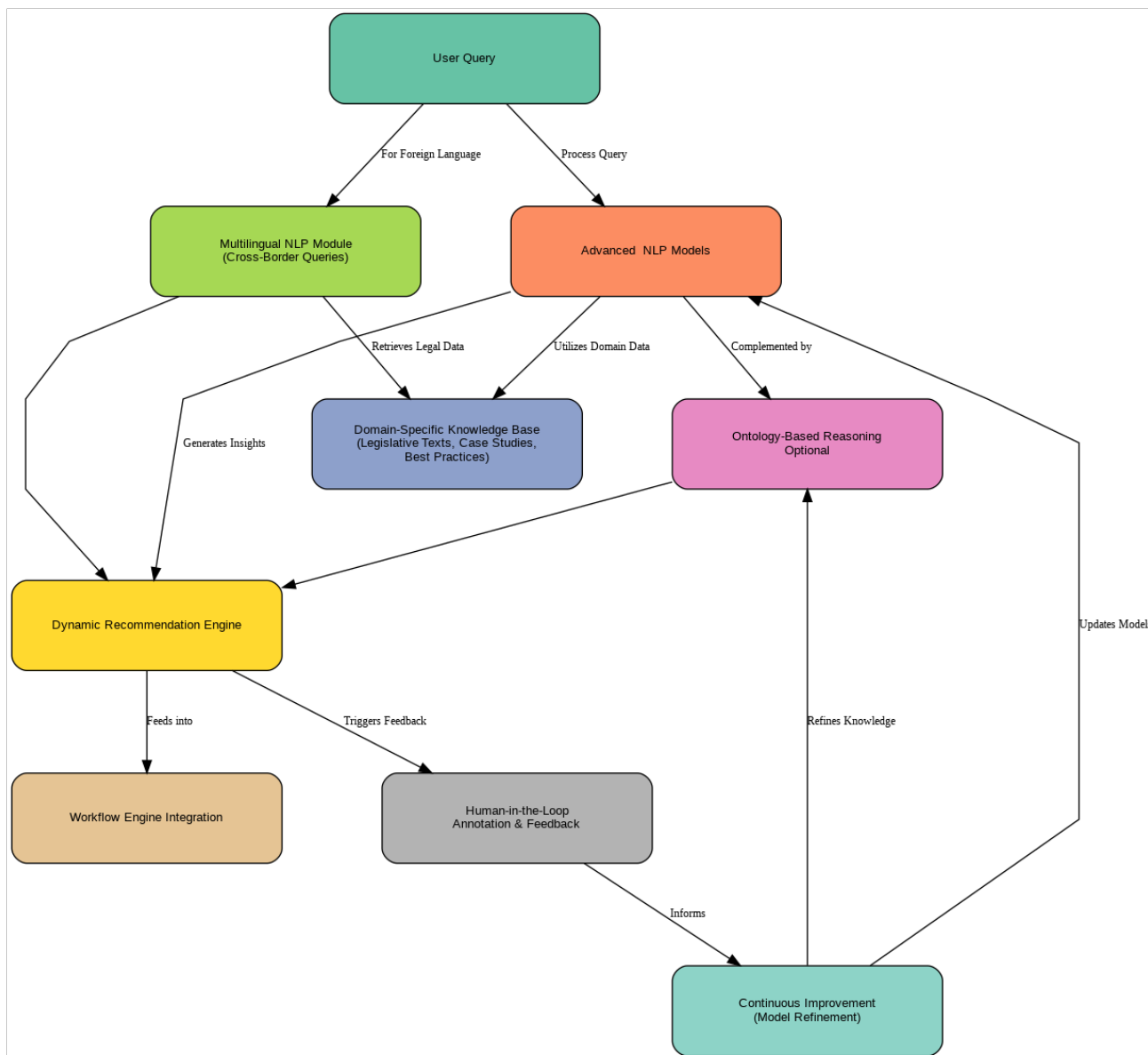


Fig. 4: Architecture of the Intelligent Interface

3. Integrated Component Interactions

This section outlines how the core components interact and flow within the system to deliver a seamless monetary claim procedure.

We begin with the BPM engine orchestrating the claim process overall, which governs the end-to-end workflow. When a claim is initiated through the Client Application, the BPM framework triggers a predefined series of steps. These steps include data capture, initial validation, and routing the claim to the appropriate processing modules.

Next, the Intelligent Processing Module comes into play with the BPM engine hands off the claim; this module collects real-time data and applies advanced ML, NLP, and LLM algorithms, preprocessing the input, extracting relevant features, and conducting risk assessments and generating recommendations and updates the BPM engine with real-time decision support—ensuring that each claim is evaluated consistently according to pre-established business rules.

Following intelligent evaluation, the system relies on the Blockchain Store to secure the transaction history. Every significant action—whether it's creating the claim record, digitally signing documents, or executing automated workflows via smart contracts—should be recorded on an immutable ledger. This ensures that all the events are transparently verifiable and reinforces the overall trustworthiness of the process.

Simultaneously, the Document Management System (DMS) handles all related documentation as users interact with the Client Application. It also cross-verifies the digital signatures and documents, ensuring that every document is authentic and tamper-proof.

These interfaces allow for secure data exchange via RESTful APIs, ensuring the system seamlessly communicates with banks, payment processors, and third-party applications.

Overall, the internal flow is a coordinated sequence where the BPM engine orchestrates the process, the Intelligent Processing Module analyses and provides decision support, the Blockchain Store secures the audit trail, and the DMS manages document authenticity. External and additional integrations could ensure that all these processes operate within a secure, interoperable, and scalable environment, resulting in an efficient, transparent, and reliable monetary claim procedure.

3.1 Workflow Orchestration and Intelligent Processing

The BPM framework is the cornerstone of the claim process, ensuring that every stage—from the initial submission to the final resolution—is executed in a structured and automated manner. When a claim is initiated via the Client Application, the BPM engine immediately captures and logs the submission, assigning it a unique identifier and triggering a predefined workflow.

Once the claim enters the workflow, it is automatically funnelled into the Intelligent Processing Module. This module leverages real-time data analytics to perform several critical functions:

- **Data Ingestion and Preprocessing:**
The module collects incoming claim data and preprocesses it by cleaning and normalising the information, ensuring that the data is in a format suitable for further analysis.
- **Risk Assessment and Classification:**
Using a combination of machine learning algorithms and rule-based models, the module classifies the claim according to its type and complexity. It assesses the associated risk—identifying potential fraud, detecting anomalies, and estimating processing delays—and flags claims that require additional review.

- **Automated Decision Support:**

Based on the processed data, the module generates insights and recommendations. These automated suggestions might include decisions such as whether to approve it

The feedback loop between the Intelligent Processing Module and the BPM engine is critical. Once the module processes a claim, it sends detailed results back to the BPM engine, which then updates the claim's status within the workflow. This update triggers subsequent automated actions, such as notifying relevant stakeholders, advancing the claim to the next processing stage, or initiating further verification steps.

In essence, this tightly integrated process ensures that claims are managed efficiently and consistently, reducing manual intervention while maintaining high accuracy and compliance. The combination of BPM orchestration and intelligent processing streamlines the claim lifecycle and provides a scalable foundation that can adapt to evolving business and regulatory requirements.

3.2 Document Management and Verification Workflow

The Document Management System (DMS) is a secure and centralised repository for all claim-related documentation. When a document is submitted via the Client Application, it undergoes a series of security measures: it is first encrypted, then digitally signed, and finally stored in the DMS. This ensures the document remains confidential, tamper-proof, and fully traceable throughout its lifecycle.

Once stored, the DMS interacts seamlessly with the Blockchain Store. In this step, the system cross-verifies each document's digital signature and hash against the immutable records maintained on the blockchain. This verification process guarantees that every document's integrity is preserved and any unauthorised modifications can be immediately detected.

In parallel, the Intelligent Processing Module and the BPM engine leverage the verified document information to bolster the overall claim validation process. By incorporating authenticated document data, these modules ensure that each claim meets procedural requirements and complies with relevant regulatory standards. This integrated approach reinforces trust in the system and ensures that all necessary documentation is secure and verifiable at every stage of the claim process.

3.3 Interim conclusion

The system architecture for the monetary claim procedure platform is crafted to be a strong, secure, and scalable solution that seamlessly combines advanced intelligent processing, blockchain security, and modular components. By utilising a comprehensive Business Process Management framework, the platform skilfully manages complex workflows, ensuring that every claim is processed consistently, transparently, and efficiently. At its heart, the Intelligent Processing Module acts as the analytical powerhouse. It uses real-time data analytics and machine learning to evaluate risk, categorise claim types, and provide automated decision support. On the other hand, the composing component plays a key role by ensuring that all interactions between the various modules are smoothly coordinated, linking user actions with backend processes. Together, these elements form a unified system architecture that tackles the technical and operational challenges of monetary claim processing and lays a solid foundation that can adapt to changing regulatory and business demands.

4. Immutable Audit Trail System Using Permissioned Blockchain

4.1 Introduction

In today's complex business environments, maintaining the integrity and trustworthiness of audit trails is paramount, especially for systems handling sensitive case management and document verification workflows. Traditional database-driven audit logs suffer from several inherent weaknesses: they can be tampered with, lack strong verification mechanisms, and often fail to provide conclusive evidence of data integrity. This chapter introduces our permissioned blockchain solution designed specifically to enhance the audit trail capabilities within our workflow orchestration and case management system.

4.2 Architectural Integration

As shown in the system architecture diagrams, our blockchain-based ledger sits at a critical junction between the BPM Framework (Workflow Orchestration) and the Intelligent Processing Module. It serves as an immutable audit trail that records all significant events, actions, and decision points within case lifecycle management.

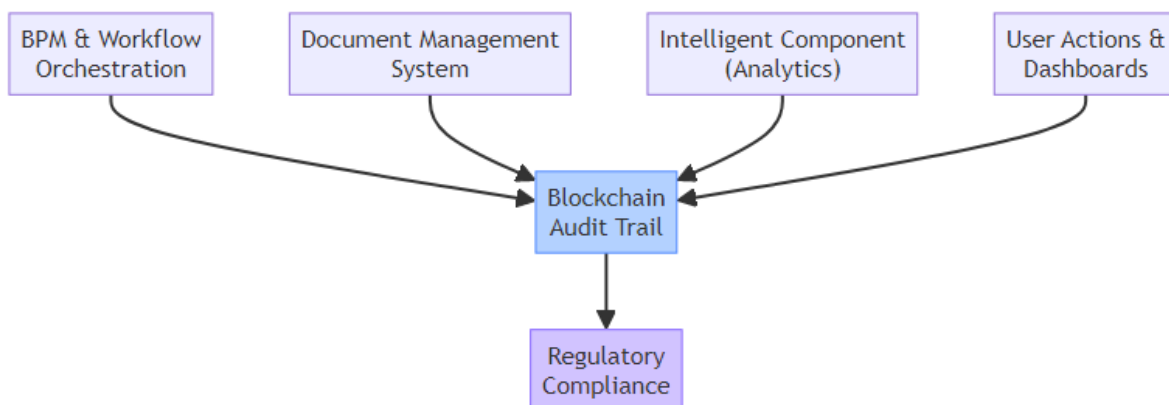


Fig. 4.1: Blockchain System Architecture showing integration layers with existing components

The permissioned blockchain component will be seamlessly integrated with the Audit Trail Module depicted in Figure 4.1, where it currently "Logs Status & Results" and "Maintains Compliance." Instead of relying on traditional database storage for these logs, the blockchain system will provide an immutable ledger that cryptographically guarantees the integrity of all recorded information. This extends to external verification flows, where the blockchain creates a verifiable record of all interactions with external systems and data sources.

Furthermore, the Document Management System will benefit from blockchain integration as document verification results and chain-of-custody information becomes permanently recorded with tamper-evident properties. When users perform critical actions, make decisions, or complete tasks, these events will be captured with non-repudiation guarantees, ensuring accountability throughout the entire process.

4.3 Technical Design of the Permissioned Blockchain

4.3.1 Consensus Mechanism

Our blockchain implementation uses a Byzantine Fault Tolerant (BFT) consensus protocol without proof-of-work mining. Specifically, we've implemented a Byzantine Fault Tolerance variant optimized for permissioned environments where all participants are known entities.

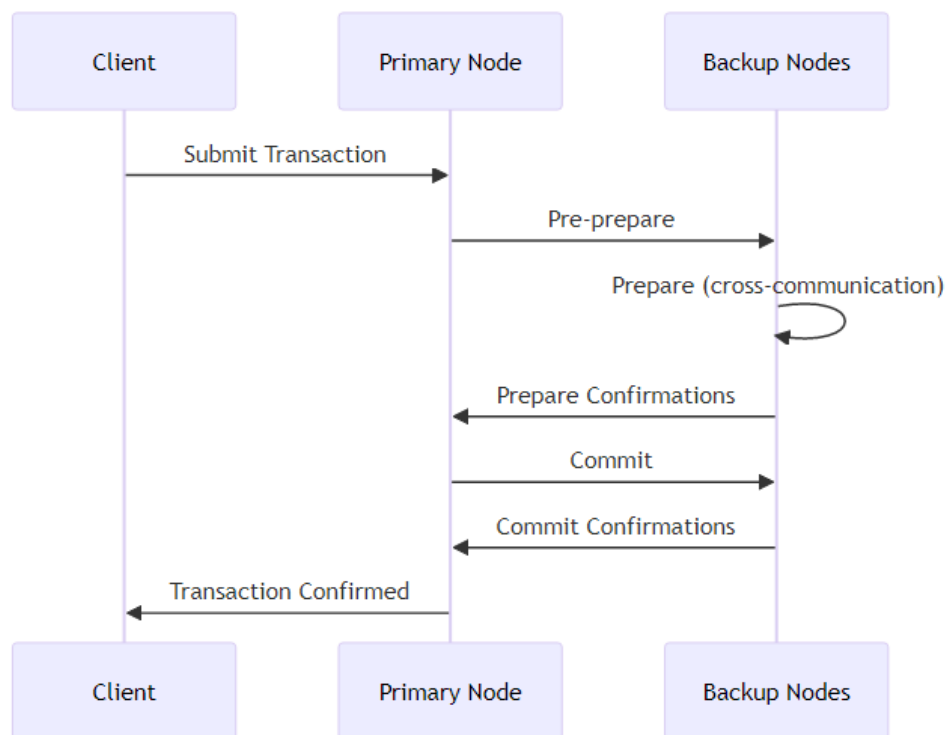


Fig. 4.2: Byzantine Fault Tolerant consensus flow showing the pre-prepare, prepare, and commit phases

This approach brings several advantages to our system architecture. Transactions confirm within seconds rather than minutes, providing the low latency necessary for interactive business processes. Once transactions are confirmed, they achieve immediate finality with no possibility of chain reorganization, which eliminates uncertainty in critical business records. By avoiding computationally expensive proof-of-work calculations, our solution remains energy efficient while maintaining security through signature-based validation where authorized participants cryptographically sign all transactions and blocks.

4.3.2 Node Architecture and Federation

The blockchain operates as a federated network with distinct node types serving different functions within the ecosystem. Validator nodes represent the core of the network, operated by authorized parties from different departments or partner organizations who actively participate in transaction validation and block production. These validators collectively ensure that only legitimate transactions enter the permanent record.

Observer nodes complement the validators by maintaining full copies of the blockchain for verification and auditing purposes without participating in consensus. These nodes enable stakeholders to independently verify the blockchain's contents without requiring validator privileges. To facilitate integration with existing systems, API Gateway nodes provide standardized REST interfaces for interaction between the blockchain and main application components.

This federation model is governed by a permissioning system that carefully controls which entities can operate validator nodes. This approach ensures consensus remains in trusted hands while still distributing trust across organizational boundaries, preventing any single entity from having unilateral control over the historical record.

4.3.3 Block and Transaction Structure

Each block in our blockchain contains a structured set of data designed to balance transparency with appropriate data protection. The block header carries essential metadata including block height, timestamp, a cryptographic digest of included transactions (Merkle root), reference to the previous block, and validator signatures that collectively endorse the block's validity.

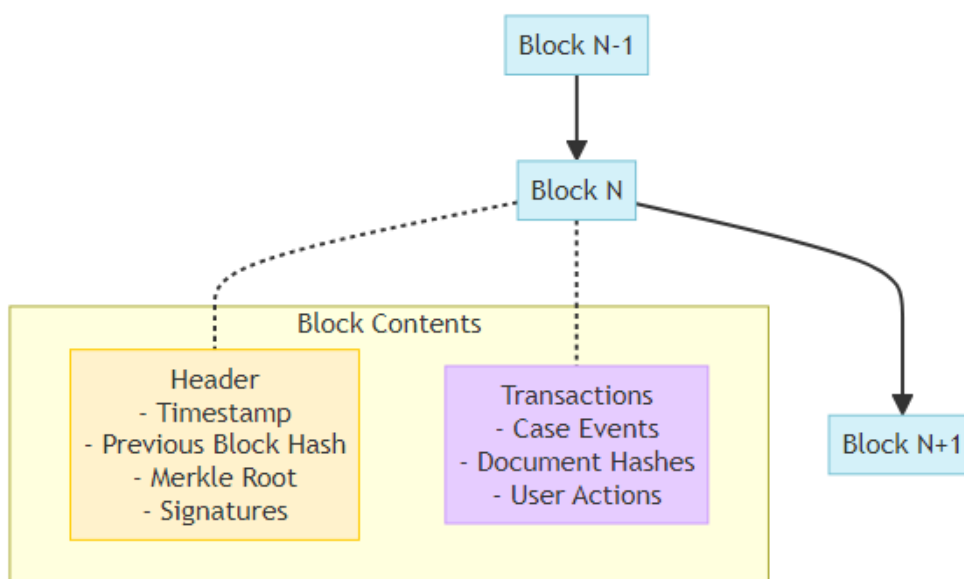


Fig. 4.3: Detailed view of block structure and transaction composition

Transaction payloads form the substantive content of each block, capturing the essential context and details of business events. Each transaction references relevant case identifiers, records the actor (whether human user or system component), specifies the action type, includes precise timestamps, and carries appropriate cryptographic signatures. The actual business data may be stored directly, hashed, or encrypted depending on sensitivity requirements.

Additional metadata accompanies these transactions to support compliance objectives, including tags that facilitate regulatory reporting, indicators for data retention policies, and access control information that governs visibility of sensitive records.

4.4 Integration Points with Existing System Components

4.4.1 BPM Engine Integration

The BPM Engine serves as a central orchestration component in our architecture, interpreting standardized BPMN process definitions and executing the corresponding tasks. We extend this engine to commit significant workflow state transitions to the blockchain at critical junctures in the process lifecycle.

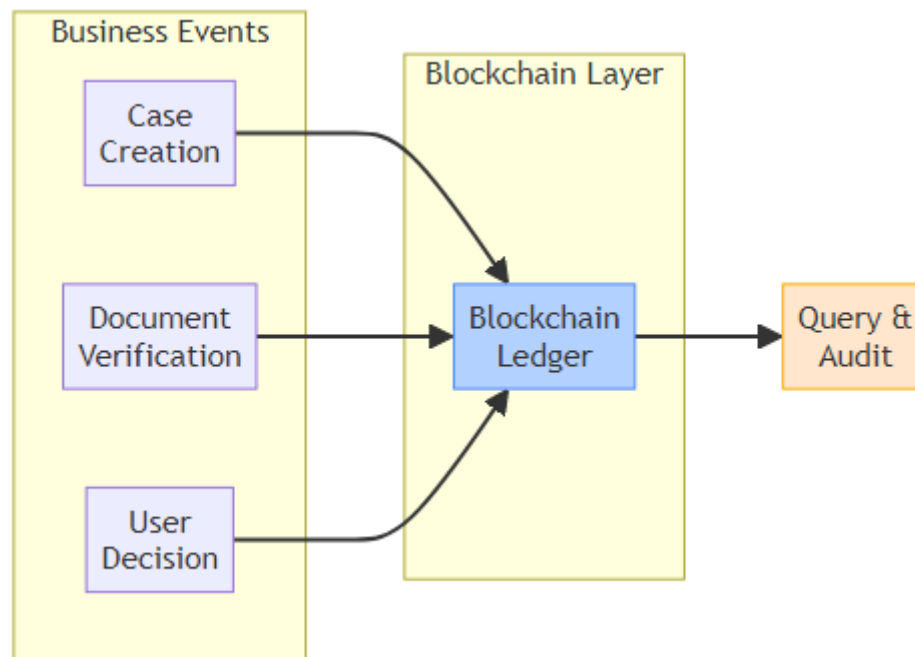


Fig. 4.4: Workflow showing how case lifecycle events are recorded to the blockchain

When workflow instances are created, modified, or completed, the engine will generate corresponding blockchain transactions that permanently record these state changes. Similarly, the assignment and completion of human tasks trigger blockchain records that establish clear accountability for who performed which actions and when. At decision points within processes, especially those with compliance implications, the decision logic, inputs, and outcomes are recorded to provide full auditability of the decision-making process.

4.4.2 Document Verification Integration

Document verification represents another crucial integration point for blockchain technology. The authenticity and integrity of documents often form the foundation of regulatory compliance in many industries. Our blockchain system creates an unalterable record each time a document undergoes verification.

This record captures the document's cryptographic hash (rather than the document itself), the verification outcome, detailed verification attributes, and precise timing information. By recording document hashes rather than full content, we maintain privacy while still enabling any party to later verify that a given document matches what was previously processed and approved.

4.4.3 Intelligent Component Integration

The Intelligent Component, which provides analytics and decision support capabilities, interacts with the blockchain in a bidirectional manner. First, it uses the blockchain as a trusted source of historical data for analysis, drawing upon the guaranteed-authentic record of past events, decisions, and outcomes to inform its analytical models.

In the opposite direction, when the Intelligent Component generates critical insights or automated decisions with significant business or compliance implications, these outputs are recorded to the blockchain. This creates accountability even for machine-generated decisions, allowing reconstruction of how automated systems reached particular conclusions.

4.5 GDPR Compliance and Data Protection

4.5.1 The Blockchain Paradox: Immutability vs. Right to be Forgotten

The implementation of blockchain technology for audit trail purposes presents a fundamental tension with certain aspects of the General Data Protection Regulation (GDPR), particularly Article 17's "Right to Erasure" (commonly known as the "Right to be Forgotten"). This tension arises from the inherent immutability of blockchain records, which by design cannot be altered or deleted once committed to the chain.

Our architecture addresses this apparent contradiction through a carefully designed technical approach that respects both blockchain integrity and data protection requirements.

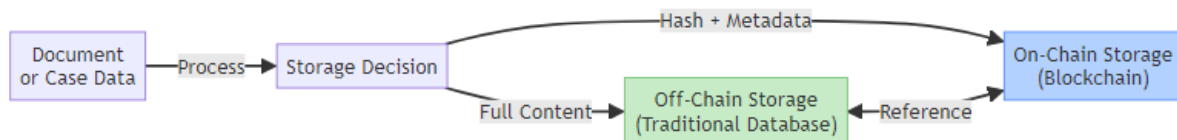


Fig. 4.5: Hybrid storage model showing on-chain vs. off-chain data storage with pseudonymization

4.5.2 Data Minimization and Pseudonymization

Our foundational approach to GDPR compliance centers on strict data minimization principles. The blockchain itself never stores personal data directly. Instead, we implement a comprehensive pseudonymization strategy whereby:

Personal identifiers are replaced with pseudonymous identifiers before any record enters the blockchain. These pseudonyms are generated through cryptographic one-way functions, making re-identification computationally impractical without access to separately stored mapping tables.

Document contents and personal data fields remain in conventional databases with appropriate access controls and encryption, while only cryptographic hashes and reference pointers appear on the blockchain itself.

Transaction metadata is carefully filtered to exclude any elements that might constitute personal data under GDPR definitions, with particular attention to possible inference attacks.

4.5.3 Technical Implementation of the Right to be Forgotten

To address the right to erasure while preserving blockchain integrity, we've implemented a cryptographic approach to functional erasure:

1. ****Key Destruction Technique****: Personal data references are encrypted using a key-based system where deletion of the encryption key renders the reference permanently inaccessible while maintaining the hash integrity of the blockchain.
2. ****Off-Chain Storage with Blockchain Pointers****: The architecture maintains a clear separation between the immutable blockchain (containing only hashes and non-personal identifiers) and conventional databases (containing actual personal data that can be properly erased).
3. ****Practical Erasure Process****: When a data subject exercises their right to erasure, the system:
 - Deletes all personal data from conventional databases
 - Removes the encryption keys needed to link blockchain references to the individual
 - Documents the erasure action with an appropriate privacy-preserving record

This approach achieves "erasure" in the functional sense required by GDPR while preserving the cryptographic integrity of the blockchain structure itself.

4.5.4 Data Controller and Processor Responsibilities

The permissioned nature of our blockchain implementation provides clear identification of data controllers and processors, addressing another critical aspect of GDPR compliance. The governance structure establishes:

Explicit assignment of data controller responsibilities to specific organizational entities within the federated network, with clear accountability for data protection impact assessments, consent management, and responding to data subject requests.

Data processing agreements that govern the responsibilities of all participating nodes in the permissioned network, particularly regarding security measures, breach notification protocols, and limitations on data use.

Documentation of data flows across the hybrid architecture, facilitating appropriate records of processing activities as required under Article 30.

4.5.5 Privacy by Design Elements

Beyond these specific measures, the entire blockchain audit trail system embodies privacy by design principles through:

Regular privacy impact assessments integrated into the development lifecycle, with particular attention to potential reidentification risks as the system evolves.

Default settings that implement the maximum privacy protection, requiring explicit administrative action to reduce protection levels.

Comprehensive audit mechanisms for privacy-related actions themselves, creating accountable records of who accessed personal data, when, and for what purpose.

4.6 Compliance and Regulatory Benefits

The blockchain-based audit system transforms our compliance posture through several fundamental improvements to data integrity. Traditional audit logs can be altered without detection, whereas our blockchain implementation creates tamper evidence through its cryptographic hash chains. Any attempt to modify historical records immediately breaks the chain of hashes, making tampering immediately detectable.

Non-repudiation represents another substantial benefit, as cryptographic signatures ensure that actors cannot credibly deny their recorded actions. This substantially strengthens accountability throughout the system. The distributed nature of our ledger eliminates single points of failure or corruption that exist in centralized database approaches, enhancing overall data integrity.

From a regulatory perspective, this architecture enables historical verification where auditors can independently examine the complete history of cases, documents, and decisions without relying on potentially manipulated exports or reports. The permissioned nature of our blockchain ensures that the right stakeholders have appropriate visibility while maintaining confidentiality from external parties, creating a balanced approach to transparent governance.

4.7 Implementation Approach

4.7.1 *Technology Foundation*

Our implementation builds upon existing enterprise blockchain frameworks, customized for our specific requirements around Byzantine Fault Tolerance and permissioned access. The integration layer connects application components to the blockchain through carefully designed adapters that handle the translation between business events and blockchain transactions. To protect the cryptographic keys that underpin system security, we implement robust key management with appropriate hardware security.

For interoperability and data access, we provide multiple interface options including both GraphQL for complex queries and REST for simpler integrations, giving flexibility in how various components and external systems interact with the blockchain.

4.7.2 *Data Privacy Considerations*

While blockchain provides powerful immutability and transparency, we carefully balance these attributes with essential data privacy requirements. We follow a hybrid storage model where only hashes, metadata, and non-sensitive information reside on-chain, while full documents and

personally identifiable information remain in conventional secure storage systems, linked by cryptographic references.

For particularly sensitive transaction details, we employ encryption with sophisticated key management that restricts data access to authorized parties. In specific compliance verification scenarios, we implement cryptographic protocols that allow proving compliance without revealing underlying sensitive data, providing verification without exposure.

4.7.3 Performance Considerations

The blockchain component is designed with careful attention to performance characteristics to ensure it enhances rather than constrains system responsiveness. High-frequency events are intelligently aggregated to prevent blockchain congestion while maintaining appropriate granularity for audit purposes. Historical data beyond mandatory retention periods can be archived from active nodes while preserving cryptographic verifiability, optimizing system performance. For exceptionally high-volume scenarios, the architecture supports partitioning approaches that maintain security while scaling horizontally.

4.8 Query and Reporting Capabilities

The blockchain audit trail system provides sophisticated query capabilities that support both operational needs and compliance requirements. Users can examine events within specific time periods, retrieve the complete history of particular cases, or investigate all actions performed by specific users or system components. Document lifecycle tracking reveals the complete verification and usage history of important assets, while preconfigured compliance reports address common regulatory requirements with minimal configuration effort.

4.9 Future Directions

Although our current implementation focuses on internal system integrity, the architecture supports future evolution in several directions. The system could extend to partner organizations through federated blockchain networks that maintain appropriate boundaries while enabling cross-organizational verification. Regulatory authorities might eventually receive direct but controlled access to blockchain data relevant to their oversight responsibilities. The Intelligent Component could develop enhanced capabilities for monitoring blockchain patterns to proactively identify compliance risks before they materialize.

4.10 Summary

The implementation of a permissioned, BFT-based blockchain for audit trail functionality fundamentally transforms the integrity, trustworthiness, and compliance capabilities of our case management and document verification workflows. By strategically integrating this technology with our existing BPM and intelligent processing components, we create a system that not only satisfies current regulatory demands but establishes a foundation for sustained compliance in an increasingly complex regulatory environment. The immutable nature of blockchain records, combined with non-repudiation through cryptographic signatures, creates unprecedented accountability and auditability throughout the entire business process lifecycle.

The careful design choices around data privacy, GDPR compliance, and permissioned access ensure that our blockchain implementation delivers these benefits without compromising on data protection principles or creating new regulatory risks. This balanced approach represents the future of compliant audit trail systems that protect both organizational and individual interests.

References

- [1] Khurana, Diksha, et al. "Natural language processing: state of the art, current trends and challenges." *Multimedia tools and applications* 82.3 (2023): 3713-3744.
- [2] Das, Badhan Chandra, M. Hadi Amini, and Yanzhao Wu. "Security and privacy challenges of large language models: A survey." *ACM Computing Surveys* 57.6 (2025): 1-39.
- [3] Qin, Libo, et al. "Large language models meet nlp: A survey." *arXiv preprint arXiv:2405.12819* (2024).
- [4] Roshani, Mohammad & Zhou, Xiangyu & Qiang, Yao & Suresh, Srinivasan & Hicks, Steve & Sethuraman, Usha & Zhu, Dongxiao. (2024). Generative LLM Powered Conversational AI Application for Personalized Risk Assessment: A Case Study in COVID-19 (Preprint). 10.2196/preprints.67363.
- [5] Grohs, Michael, et al. "Large language models can accomplish business process management tasks." *International Conference on Business Process Management*. Cham: Springer Nature Switzerland, 2023.
- [6] Shanmugaraja P, Tamil Nadu, Susmitha K. S; Swadha S; Vijay R; Naveen G. "Document Management System using Blockchain," 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), Trichy, India, 2022, pp. 888-891, doi: 10.1109/ICAISS55157.2022.10011076.
- [7] Hegab, Mohamed & Nassar, Khaled. (2005). Decision Support System for Commencement Delay Claims. *Practice Periodical on Structural Design and Construction*. 10. 10.1061/(ASCE)1084-0680(2005)10:3(177).
- [8] Ramadhan, Andhika & Pane, Kharisma & Wardhana, Kun & Suharjo, (2023). Blockchain and API Development to Improve Relational Database Integrity and System Interoperability. *Procedia Computer Science*. 216. 151-160. 10.1016/j.procs.2022.12.122.
- [9] Ramadhan, Andhika & Pane, Kharisma & Wardhana, Kun & Suharjo, (2023). Blockchain and API Development to Improve Relational Database Integrity and System Interoperability. *Procedia Computer Science*. 216. 151-160. 10.1016/j.procs.2022.12.122.